

NCERT CLASS 11 BIOLOGY • CHAPTER 2

Biological Classification

Complete Exam-Oriented Academic Notes, Visual Diagrams, Taxonomy Tables, Mnemonics & Solutions

NCERT TEXTBOOK

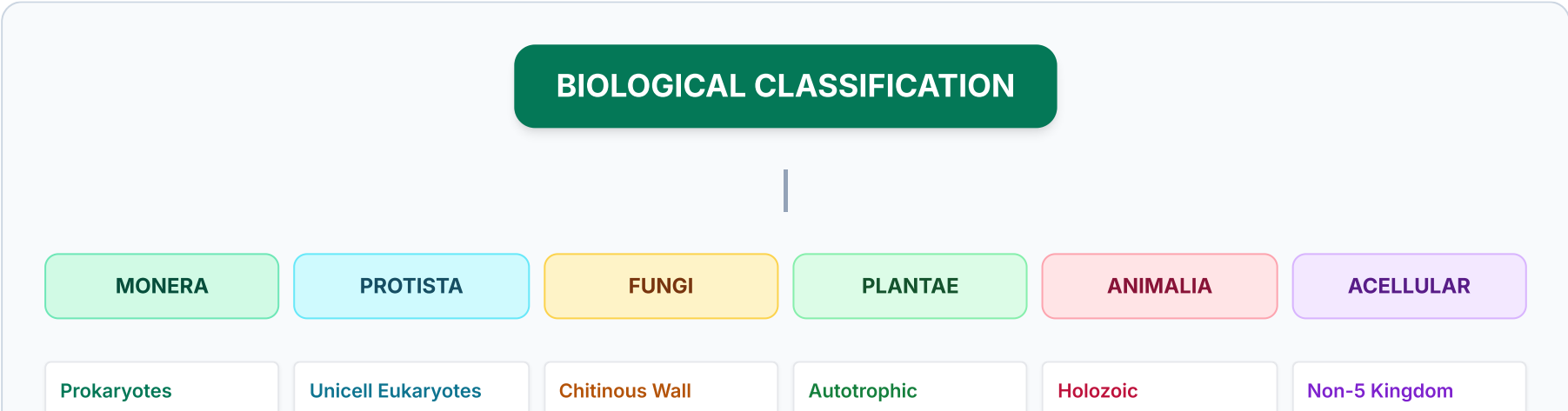
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Chapter Mind Map & Taxonomic Overview

Visual Blueprint



- Archaeobacteria
- Eubacteria
- Cyanobacteria
- Mycoplasma

- Chrysophytes
- Dinoflagellates
- Euglenoids
- Slime Moulds
- Protozoans

- Phycomycetes
- Ascomycetes
- Basidiomycetes
- Deuteromycetes

- Algae
- Bryophytes
- Pteridophytes
- Gymnosperms
- Angiosperms

- Multicellular
- No cell wall
- Heterotrophic
- Sensory System

- Viruses (RNA/DNA)
- Viroids (Free RNA)
- Prions (Proteins)
- Lichens (Symbiosis)

1. History & Systems of Classification

NCERT Pages 11–12

Early & Two-Kingdom Systems

Since the dawn of human civilization, organisms were classified instinctively based on utility—for **food, shelter, and clothing**.

Aristotle (Earliest Scientific Basis)

- **Plants:** Classified using simple morphological characters into *Trees, Shrubs, and Herbs*.
- **Animals:** Divided into two groups: those with red blood (*Enaima*) and those without red blood (*Anaima*).

Carolus Linnaeus (Two Kingdom System)

Established two kingdoms: **Kingdom Plantae** and **Kingdom Animalia**.

Demerits of Two-Kingdom System:

- Did not distinguish between **Eukaryotes** and **Prokaryotes**.
- Did not separate **Unicellular** and **Multicellular** organisms (e.g., *Chlamydomonas* and *Spirogyra* were grouped together under algae).

R.H. Whittaker's Five Kingdom System (1969)

Five Main Criteria Used by R.H. Whittaker:

1. **Cell Structure:** Prokaryotic vs. Eukaryotic nature.
2. **Body Organisation:** Cellular, Tissue, Organ, or Organ system.
3. **Mode of Nutrition:** Autotrophic (Photosynthetic/Chemosynthetic) vs. Heterotrophic (Saprophytic/Parasitic/Holozoic).
4. **Reproduction:** Asexual, Sexual, Cell fusion.
5. **Phylogenetic Relationships:** Evolutionary history and origins.

Memory Trick: Whittaker's Criteria

Remember: "**C-B-N-R-P**" (Cells Build Nutrition, Reproduce Phylogeny)

- Cell Structure | Body Organisation | Nutrition | Reproduction | Phylogeny

Three-Domain System (Six Kingdom Classification):

Divides Kingdom Monera into **two domains** (Archaea and Bacteria), leaving all eukaryotic kingdoms in the **third domain** (Eukarya).

- Grouped **Photosynthetic** green plants with **Non-photosynthetic** heterotrophic fungi (Chitinous wall vs. Cellulosic wall).

 **Table 2.1: Characteristics of the Five Kingdoms (NCERT Page 11)**

Characters	Monera	Protista	Fungi	Plantae	Animalia
Cell Type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell Wall	Non-cellulosic (Polysaccharide + amino acid)	Present in some	Present (with chitin)	Present (cellulose)	Absent
Nuclear Membrane	Absent	Present	Present	Present	Present
Body Organisation	Cellular	Cellular	Multicellular / Loose tissue	Tissue / Organ	Tissue / Organ / Organ system
Mode of Nutrition	Autotrophic (Chemosynthetic & Photosynthetic) & Heterotrophic (Saprophytic / Parasitic)	Autotrophic (Photosynthetic) & Heterotrophic	Heterotrophic (Saprophytic / Parasitic)	Autotrophic (Photosynthetic)	Heterotrophic (Holozoic / Saprophytic etc.)

 2. Kingdom Monera

NCERT Pages 12–14

Bacteria are the sole members of Kingdom Monera. They are the most abundant micro-organisms and live almost everywhere (soil, extreme hot springs, deserts, snow, deep oceans, and parasites on/in organisms).

Categories Based on Bacterial Shape (Figure 2.1):

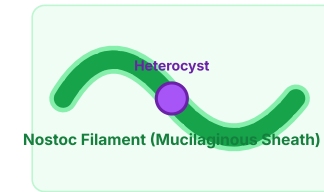
Figure 2.1 & 2.2 Diagrams

Coccus (pl. cocci)
Spherical shape

Bacillus (pl. bacilli)
Rod-shaped

Vibrium (pl. vibrio)
Comma-shaped

Spirillum (pl. spirilla)
Spiral shaped



Metabolic Diversity:

Bacteria show the **most extensive metabolic diversity** among all living organisms. They can be autotrophic (photosynthetic/chemosynthetic) or heterotrophic (vast majority).

Reproduction in Bacteria:

- Mainly by **Binary Fission**.
- Produce **Spores** under unfavourable conditions.
- Primitive sexual reproduction: **DNA transfer** from one bacterium to another.

🏠 2.1.1 Archaeobacteria

Live in harsh environments due to a **different cell wall structure**:

- **Halophiles**: Extreme salty areas.
- **Thermoacidophiles**: Hot sulfur springs.
- **Methanogens**: Marshy areas & gut of ruminants (cows/buffaloes); produce *methane (biogas)*.

⚙️ 2.1.2 Eubacteria ('True Bacteria')

Rigid cell wall, flagellum if motile.

- **Cyanobacteria (Blue-green algae)**: Chlorophyll *a*, photosynthetic. Form gelatinous sheath/blooms. Have **Heterocysts** for N_2 fixation (e.g., *Nostoc*, *Anabaena*).
- **Chemosynthetic Autotrophs**: Oxidise nitrates, nitrites, ammonia for ATP; recycle N, P, Fe, S.
- **Heterotrophic Bacteria**: Decomposers. Used for curd, antibiotics, N_2 fixation. Pathogens causing *cholera*, *typhoid*, *tetanus*, *citrus canker*.

🦠 Mycoplasma

- **Completely lack a cell wall.**
- **Smallest living cells** known.
- Can survive **without oxygen**.
- Many are pathogenic in plants & animals.

≡ 3. Kingdom Protista

NCERT Pages 14–16

Includes all **single-celled eukaryotes**. Primarily aquatic. Forms a link with plants, animals, and fungi. Protistan cell body contains a well-defined nucleus and membrane-bound organelles.

Group	Habitat & Features	Cell Wall / Outer Layer	Locomotion / Special Features	Examples
1. Chrysophytes	Freshwater & Marine, microscopic plankton, photosynthetic. Chief producers in oceans.	Two thin overlapping shells fitting as a soap box , embedded with silica (indestructible).	Passive floating in water currents. Forms Diatomaceous Earth (used in polishing, oil/syrup filtration).	Diatoms, Golden Algae (Desmids).
2. Dinoflagellates	Mostly marine & photosynthetic. Yellow, green, brown, blue, red pigments.	Stiff cellulose plates on the outer surface.	2 Flagella (one longitudinal, one transverse in wall furrow). Causes Red Tides & releases toxins.	<i>Gonyaulax</i> .
3. Euglenoids	Majority fresh water in stagnant water. Mixotrophic (Photosynthetic in sun, heterotrophic in dark).	No cell wall; protein-rich layer called Pellicle (makes body flexible).	2 Flagella (one short, one long). Pigments identical to higher plants.	<i>Euglena</i> (Fig 2.4b).
4. Slime Moulds	Saprophytic protists. Body moves along decaying twigs/leaves engulfing organic matter.	Spores possess true cellulose walls (extremely resistant, air-dispersed).	Forms aggregation called Plasmodium in suitable conditions; forms fruiting bodies in unfavourable conditions.	Slime Moulds (Fig 2.4c).
5. Protozoans	Heterotrophs (predators or parasites). Primitive relatives of animals. Divided into 4 groups.	Silica shells in marine amoeboids; others lack walls.	Pseudopodia, Flagella, Cilia, or Spore-stage.	<i>Amoeba</i> , <i>Trypanosoma</i> , <i>Paramecium</i> , <i>Plasmodium</i> .

Amoeboid Protozoans:

Move & capture prey using **pseudopodia (false feet)**. Marine forms have silica shells. Parasite: *Entamoeba*.

Flagellated Protozoans:

Free-living or parasitic. Possess free flagella. Cause diseases like **Sleeping Sickness** (*Trypanosoma*).

Ciliated Protozoans:

Aquatic, actively moving with thousands of **cilia**. Has a cavity (**gullet**). E.g., *Paramecium*.

Sporozoans:

Infectious spore-like stage in life cycle. Most notorious: *Plasmodium* (malarial parasite causing malaria).

4. Kingdom Fungi

NCERT Pages 16–18

General Characteristics & Structure

Unique kingdom of **heterotrophic organisms**. Cosmopolitan (air, water, soil, animals, plants). Prefer warm and humid places.

- **Morphology:** Except for *Yeast* (unicellular), fungi are **filamentous**.
- **Hyphae & Mycelium:** Long, slender thread-like structures called **hyphae**. The network of hyphae is called **mycelium**.
- **Coenocytic Hyphae:** Continuous tubes filled with multinucleated cytoplasm (aseptate).
- **Septate Hyphae:** Have cross-walls or septae.
- **Cell Wall Composition:** Composed of **Chitin** and **Polysaccharides**.
- **Nutrition:** Saprophytes (dead matter), Parasites (living hosts), or Symbionts (Lichens with algae; Mycorrhiza with plant roots).

Reproduction in Fungi

- 1. **Vegetative Means:** Fragmentation, Fission, Budding.
- 2. **Asexual Spores:** Conidia, Sporangiospores, Zoospores.
- 3. **Sexual Spores:** Oospores, Ascospores, Basidiospores (produced in *fruiting bodies*).

Three Steps of Fungal Sexual Cycle:

- 1. **Plasmogamy:** Fusion of protoplasts between two motile or non-motile gametes.
- 2. **Karyogamy:** Fusion of two nuclei.
- 3. **Meiosis:** Reduction division in zygote producing haploid spores.

Dikaryon / Dikaryophase (\$n+n\$ condition): Occurs in Ascomycetes & Basidiomycetes where nuclear fusion is delayed.

Comparative Breakdown of the Four Classes of Kingdom Fungi

Class	Habitat	Mycelium	Asexual Reproduction	Sexual Reproduction & Spores	Examples & Notes
Phycomycetes	Aquatic habitats, decaying wood, obligate plant parasites.	Aseptate & Coenocytic	Zoospores (motile) or Aplanospores (non-motile); endogenously in sporangium.	Zygospore formed by gamete fusion (Isogamous/Anisogamous/Oogamous).	<i>Mucor</i> , <i>Rhizopus</i> (bread mould), <i>Albugo</i> (parasitic on mustard).
Ascomycetes (Sac-Fungi)	Saprophytic, decomposers, parasitic, coprophilous (dung).	Branched & Septate	Conidia produced exogenously on special <i>conidiophores</i> .	Ascospores produced endogenously in sac-like Asci inside Ascocarps .	<i>Penicillium</i> , <i>Yeast</i> (unicell), <i>Aspergillus</i> , <i>Claviceps</i> , <i>Neurospora</i> (genetics), Morels & Truffles (edible).

Class	Habitat	Mycelium	Asexual Reproduction	Sexual Reproduction & Spores	Examples & Notes
Basidiomycetes (Club-Fungi)	Soil, logs, tree stumps, parasites (rusts/smuts).	Branched & Septate	Asexual spores generally not found; fragmentation common.	Sex organs absent; Plasmogamy by somatic fusion $\rightarrow$ Basidiospores exogenously on Basidium inside Basidiocarps .	<i>Agaricus</i> (mushroom), <i>Ustilago</i> (smut), <i>Puccinia</i> (rust fungus).
Deuteromycetes (Fungi Imperfecti)	Decomposers of litter, mineral cycling, saprophytes/parasites.	Branched & Septate	Reproduce only by asexual spores known as Conidia .	Sexual stage unknown/absent. (Moved to Ascomycetes/Basidiomycetes when sexual stage discovered).	<i>Alternaria</i> , <i>Colletotrichum</i> , <i>Trichoderma</i> .



Memory Trick: Classes of Fungi & Examples

Classes Mnemonic: "P-A-B-D" (Play All Best Dances)

Phycomycetes, Ascomycetes, Basidiomycetes, Deuteromycetes.

Phycomycetes Examples: "RAM"

Rhizopus, Albugo, Mucor.

🌲 5. Kingdoms Plantae & Animalia

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2.4 Kingdom Plantae

Includes eukaryotic chlorophyll-containing organisms (plants) with cellulosic cell walls.

Partially Heterotrophic Plants:

- **Insectivorous Plants:** *Bladderwort* and *Venus fly trap*.
- **Parasitic Plant:** *Cuscuta*.

Groups Included: Algae, Bryophytes, Pteridophytes, Gymnosperms, Angiosperms.



2.5 Kingdom Animalia

Heterotrophic eukaryotic, multicellular organisms whose cells **lack cell walls**.

- **Nutrition: Holozoic** (by ingestion of food). Reserve food stored as **glycogen or fat**.
- **Growth Pattern:** Follow a definite growth pattern into adults with definite shape and size.
- **Mechanisms:** Higher forms show elaborate **sensory and neuromotor mechanisms**.
- **Locomotion:** Most are capable of locomotion.

Life Cycle & Alternation of Generations: Alternation between diploid sporophytic ($2n$) and haploid gametophytic (n) phases.

• **Reproduction:** Sexual reproduction by copulation of male and female followed by embryological development.

6. Acellular Entities & Lichens

NCERT Pages 19–21

Note: Viruses, Viroids, Prions, and Lichens were NOT included in R.H. Whittaker's 5 Kingdom Classification system because viruses/viroids/prions are acellular/non-cellular entities.

Viruses (Inert Crystalline Structures)

Non-cellular organisms characterized by having an inert crystalline structure outside living cells. Once inside a host, they take over cellular machinery to replicate, killing the host. They are **obligate parasites**.

Key Historical Discoveries:

Dmitri Ivanowsky (1892): Discovered Tobacco Mosaic Virus (TMV); smaller than bacteria as they passed through bacteria-proof filters.

M.W. Beijerinck (1898): Called infected fluid "*Contagium vivum fluidum*" (infectious living fluid) and coined the term "Virus" (venom/poisonous fluid).

W.M. Stanley (1935): Showed viruses could be **crystallized**, consisting largely of proteins.

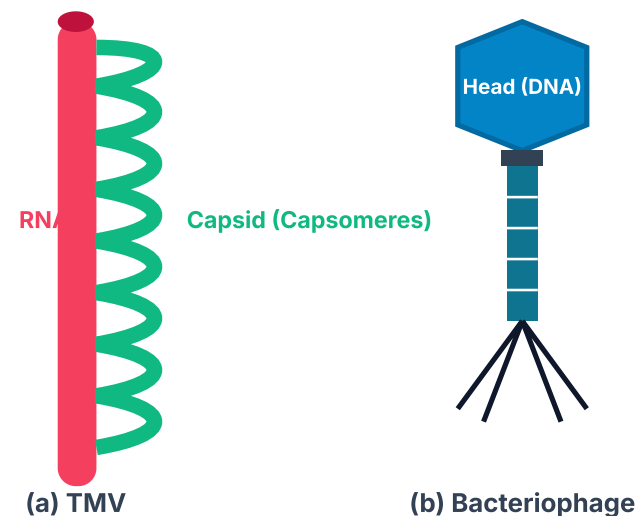
Structure & Genetic Material:

Viruses are **nucleoproteins** (Protein coat called **Capsid** made of **Capsomeres** + Nucleic Acid).

⚠ NO VIRUS CONTAINS BOTH RNA AND DNA!

- **Plant Viruses:** Usually Single-Stranded RNA (ssRNA).
- **Animal Viruses:** Single or Double-Stranded RNA, or Double-Stranded DNA.

Figure 2.6 (a) TMV & (b) Bacteriophage Structure



- **Bacteriophages:** Usually Double-Stranded DNA (dsDNA).

🕒 Viroids

Discovered by **T.O. Diener (1971)**.

- Smaller than viruses.
- Consists of **free RNA without a protein coat**.
- Low molecular weight RNA.
- Causes **Potato Spindle Tuber Disease**.

☠️ Prions

Infectious agents consisting of **abnormally folded protein** (similar in size to viruses).

- Causes **Bovine Spongiform Encephalopathy (BSE)** / Mad Cow Disease in cattle.
- Causes **Cr-Jacob Disease (CJD)** variant in humans.

🌿 Lichens

Symbiotic (mutually useful) association between Algae and Fungi:

- **Phycobiont:** Algal component (autotrophic; prepares food).
- **Mycobiont:** Fungal component (heterotrophic; provides shelter, absorbs water/minerals).
- **Pollution Indicators:** Do NOT grow in polluted areas (SO₂ sensitive).

🎓 7. NCERT Textbook Exercise Solutions

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Q1. Discuss how classification systems have undergone several changes over a period of time?

Answer: Early classification was based on human utility (food, shelter, clothing). Aristotle introduced simple morphological criteria (plants into trees/shrubs/herbs; animals into red-blooded vs non-red-blooded). Linnaeus developed the *Two-Kingdom system (Plantae & Animalia)*, which failed to distinguish prokaryotes from eukaryotes, unicellular from multicellular, and autotrophs from heterotrophs. As scientific understanding advanced, criteria included cell structure, cell wall nature, nutrition, reproduction, and phylogenetic relationships. This led R.H. Whittaker (1969) to propose the *Five-Kingdom Classification* (Monera, Protista, Fungi, Plantae, Animalia), and later Carl Woese proposed the *Three-Domain System*.

Q2. State two economically important uses of: (a) heterotrophic bacteria (b) archaebacteria

(a) Heterotrophic Bacteria:

- Production of curd and cheese from milk (e.g., *Lactobacillus*).
- Production of antibiotics and nitrogen fixation in legume roots (e.g., *Rhizobium*).

(b) Archaebacteria:

- Methanogens are used in biogas production (methane) from animal dung.

- Thermoacidophiles and Halophiles are source enzymes for biotechnology processes requiring extreme heat or salinity.

Q3. What is the nature of cell-walls in diatoms?

Answer: In diatoms, the cell wall forms two thin overlapping shells that fit together like a **soap box**. The walls are embedded with **silica**, making them indestructible. Over billions of years, their cell wall deposits accumulate on the ocean floor to form **diatomaceous earth**.

Q4. Find out what do the terms 'algal bloom' and 'red-tides' signify.

Answer:

- **Algal Bloom:** Rapid multiplication and excessive growth of cyanobacteria (blue-green algae) in polluted water bodies, causing discoloration and depletion of oxygen.
- **Red Tides:** Rapid multiplication of red dinoflagellates (e.g., *Gonyaulax*) causing the sea water to appear red. Toxins released by them can kill large numbers of marine animals such as fish.

Q5. How are viroids different from viruses?

Feature	Viruses	Viroids
Protein Coat (Capsid)	Present (made of capsomeres)	Absent (Free RNA)
Genetic Material	Either RNA or DNA	Only low molecular weight RNA
Size	Larger than viroids	Smaller than viruses

Q6. Describe briefly the four major groups of Protozoa.

Answer: (1) **Amoeboid:** Move via pseudopodia (e.g., *Amoeba*, *Entamoeba*). (2) **Flagellated:** Possess flagella, cause sleeping sickness (e.g., *Trypanosoma*). (3) **Ciliated:** Aquatic, move with thousands of cilia, possess gullet (e.g., *Paramecium*). (4) **Sporozoans:** Have infectious spore-like stage in life cycle (e.g., *Plasmodium*).

Q7. Plants are autotrophic. Can you think of some plants that are partially heterotrophic?

Answer: Yes. Insectivorous plants like **Bladderwort** and **Venus fly trap** capture insects for nitrogen requirement while photosynthesizing. **Cuscuta** is a completely parasitic plant lacking chlorophyll.

Q8. What do the terms phycobiont and mycobiont signify?

Answer: In lichens: **Phycobiont** refers to the algal component, which is autotrophic and synthesizes food. **Mycobiont** refers to the fungal component, which is heterotrophic, providing shelter and absorbing water/minerals for the algal partner.

Q9. Give a comparative account of the classes of Kingdom Fungi under (i) mode of nutrition (ii) mode of reproduction.

Answer: Refer to the Fungi Comparative Table in Section 4 above for a comprehensive breakdown of Phycomycetes, Ascomycetes, Basidiomycetes, and Deuteromycetes.

Q10. What are the characteristic features of Euglenoids?

Answer: Fresh water, stagnant water organisms. Lack cell wall; have a protein-rich flexible layer called **Pellicle**. Have 2 flagella (short & long). Mixotrophic: photosynthetic in sunlight, heterotrophic predators in darkness. Pigments identical to higher plants. Example: *Euglena*.

Q11. Give a brief account of viruses with respect to their structure and nature of genetic material. Also name four common viral diseases.

Answer: Viruses are acellular nucleoproteins consisting of a protein coat (capsid made of capsomeres) protecting genetic material (either RNA or DNA, never both). Plant viruses have ssRNA; animal viruses have ssRNA, dsRNA, or dsDNA; bacteriophages have dsDNA. Common viral diseases: **Mumps, Smallpox, Herpes, Influenza, AIDS.**

Q12. Organise a discussion in your class on the topic – Are viruses living or non-living?

Answer: Viruses represent a border line between living and non-living:

- **Non-Living Features:** Inert crystalline structure outside living host, lack cellular machinery, cannot reproduce independently, crystallizable.
- **Living Features:** Possess genetic material (RNA/DNA), take over host cell machinery to replicate, undergo mutation, cause diseases.

Interactive Self-Assessment Quiz

[Test Your Knowledge](#)

1. Which organism is known as the chief producer in the oceans?

A) Dinoflagellates

B) Diatoms (Chrysophytes)

C) Euglenoids

D) Cyanobacteria

2. Organisms lacking a cell wall completely and capable of surviving without oxygen are:

A) Methanogens

B) Mycoplasma

C) Slime Moulds

D) Cyanobacteria

3. Who demonstrated that the extract of infected tobacco plants could cause infection in healthy plants and named the fluid 'Contagium vivum fluidum'?

A) Dmitri Ivanowsky

B) M.W. Beijerinck

C) W.M. Stanley

D) T.O. Diener

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